

Describe a Declared Threshold Fixture

Name every promise before interpreting a reading

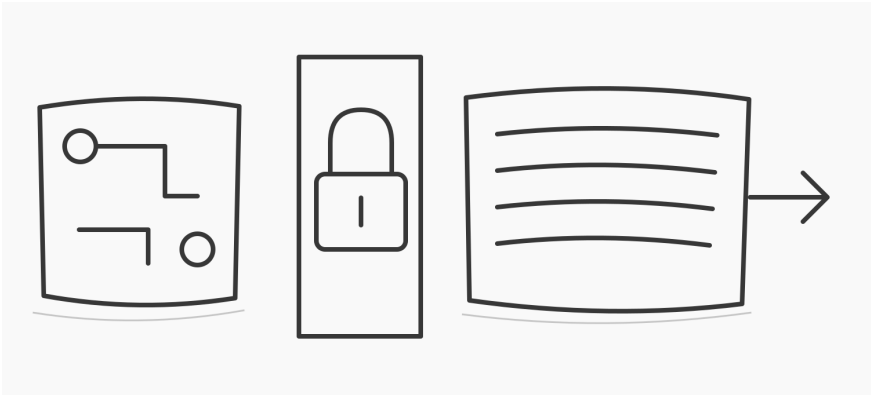
E0 COPIED REPLAY — NO UNIDENTIFIED MODULE MAY BE POWERED

Lesson	070 — component	Time	150–210 minutes
Prerequisite	copied values and explicit status	Board	compile-only Mega replay
Input	declared descriptor and frame	Observation	named validation cells

Responsibility. `ModuleThresholdDescriptor` states the copied electrical declaration for one declared fixture reference. Stateless functions validate its internal structure, validate one correlated copied `ModuleThresholdFrame`, and interpret comparator level only under declared polarity. They do not identify, probe, energize, sample, calibrate, or qualify a physical board.

1 Require identity before traits

Many boards expose an analog output, a comparator output, or both. Similar connectors and seller names do not make their voltage, pinout, polarity, output stage, pull, adjustment, or timing interchangeable. Lesson 070 begins with nonzero descriptor, declared-specimen, declared-electrical-evidence, and revision fields rather than a universal “sensor module” guess.



Alternative text: an anonymous board silhouette stops before a locked identity gate while a declared fixture card carrying complete revision references proceeds to structural validation.

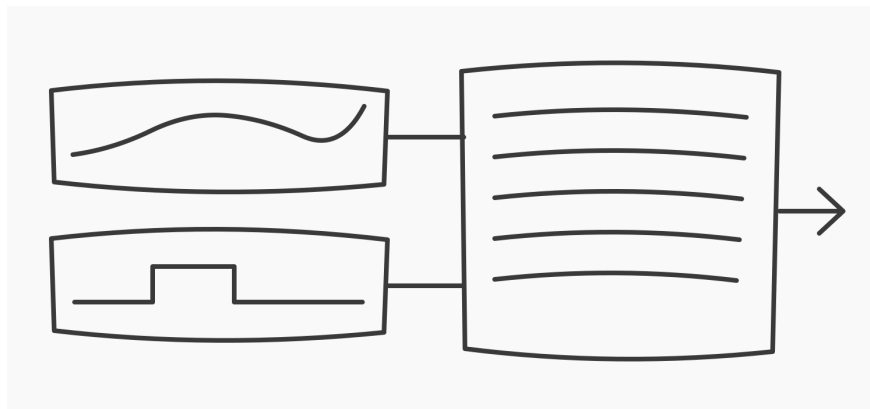
E0 can establish	E0 cannot establish
supported schema and closed values	that a connected board matches the declaration
internally coherent copied fields	voltage safety, pin order, or physical pull
descriptor/frame correlation	measured warm-up, settling, or pot direction
deterministic comparator interpretation	sensor meaning, accuracy, or suitability

2 Predict before validation

Would PCB color or a KY- alias establish an electrical identity? _____. Name the three revisions that keep a declaration attributable: _____.

3 Keep the two channel contracts separate

`ModuleChannelTopology` is `AnalogOnly`, `ComparatorOnly`, or `AnalogAndComparator`. The analog lane declares supply and signal millivolt ranges plus a raw domain. The comparator lane declares output stage, pull requirement and rail, polarity, threshold control kind, and direction. One lane never supplies a missing fact for the other.



Alternative text: separate graphite lanes carry a continuous analog trace and discrete comparator levels into one descriptor card; neither lane is inferred from the other.

Declared type	Closed vocabulary
<code>ModuleComparatorOutputStage</code>	<code>Unspecified</code> , <code>PushPull</code> , <code>OpenDrain</code> , <code>OpenCollector</code>
<code>ModulePullRequirement</code>	<code>Unspecified</code> , <code>None</code> , <code>PullUp</code> , <code>PullDown</code>
<code>ModuleDeclaredRail</code>	<code>Unspecified</code> , <code>Ground</code> , <code>LogicSupply</code> , <code>ModuleSupply</code>
<code>ModuleThresholdControlKind</code>	<code>Unspecified</code> , <code>Fixed</code> , <code>Potentiometer</code>
<code>ModuleThresholdDirection</code>	<code>Unspecified</code> , <code>IncreasingClockwise</code> , <code>IncreasingCounterclockwise</code>

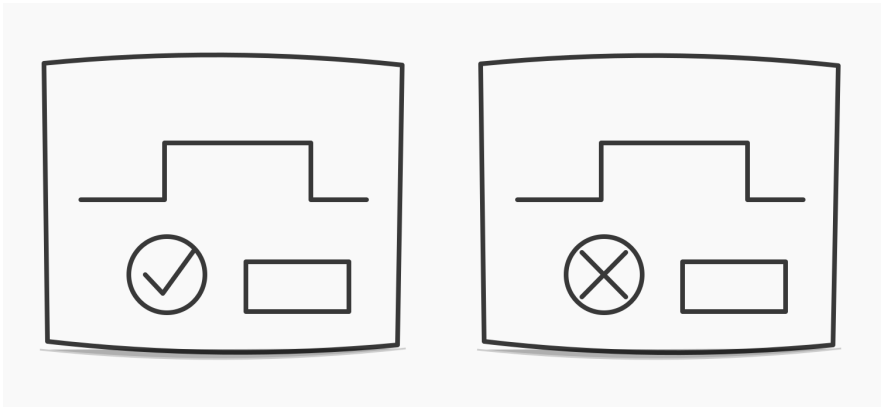
`Unspecified` is a valid E0 declaration, not permission to choose a plausible default. Invalid underlying enum encodings reject. A pull rail is unspecified when no pull is declared and must be specified when a pull is declared. Unused fields take their canonical values.

4 Treat ranges as declarations

Ordered `ModuleMillivoltRange` and `ModuleRawDomain` fields describe copied claims. Structural acceptance does not prove a rating or authorize an ADC reference. Raw endpoints are codes; Lesson 070 does not call them open, shorted, or saturated.

5 Interpret comparator level only under polarity

`moduleComparatorAsserted(descriptor, comparatorLevelHigh)` returns a `Result<bool>`. With `ActiveHigh`, a high copied level is asserted. With `ActiveLow`, it is not. `Unspecified` polarity returns failure rather than guessing.



Alternative text: the same copied low-to-high transition appears twice; an active-high declaration maps the high level to asserted, while an active-low declaration maps it to not asserted, shown with check and cross marks rather than color.

Polarity	copied level high	copied level low
ActiveHigh	asserted	not asserted
ActiveLow	not asserted	asserted
Unspecified	failure	failure

6 Bind the whole copied frame

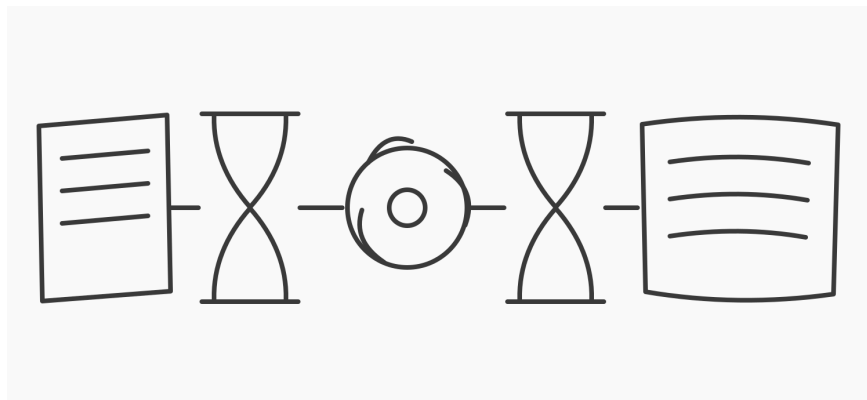
`ModuleThresholdFrame` repeats descriptor correlation and declared specimen/electrical provenance, then carries source identity, source configuration revision, sequence, supplied observation time, analog value and status, comparator level/presence/assertion and status, producer statuses, and copied declared-duration satisfaction.

`ModuleChannelStatus` distinguishes `NotPresent`, `Current`, `Stale`, and `ProducerFault`. Analog-only frames use the canonical comparator-absent representation; comparator-only frames use the canonical analog-absent representation. Presence, status, producer status, and derived assertion must agree with topology.

If analog producer status fails while the copied raw value happens to lie in range, the channel status must be _____.
Why may the raw value not repair the failure? _____.

7 Declare time; never hide a clock

`ModuleDeclaredDuration` pairs `Known` or `Unknown` with a `Duration`. `Known` zero means the declaration explicitly requires no delay. `Unknown` is canonically paired with numeric zero and blocks declaration completeness. The frame's copied warm-up or settling satisfaction must be false when the corresponding duration is unknown.



Alternative text: a declared fixture card passes a warm-up hourglass, an adjustment gesture, a settling hourglass, and finally reaches an admitted copied frame; all timing satisfaction remains supplied evidence.

Lesson 070 reads no clock, waits for nothing, and does not turn a potentiometer. The frame's warm-up-satisfied and settling-satisfied fields are copied declarations bound to its exact provenance. Lesson 071 will apply supplied time to a characterization run.

Declaration	Numeric value	Copied satisfied flag
known, zero	zero	may be true or false as correlated evidence
known, nonzero	declared duration	may be true or false as correlated evidence
unknown	canonical zero	must be false

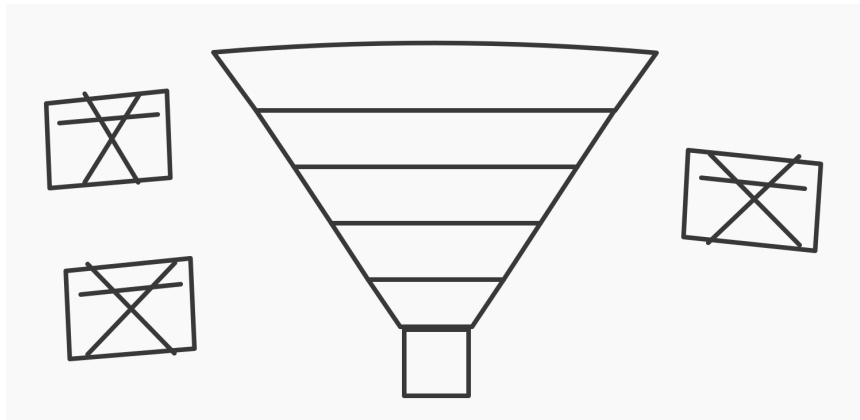
8 Do not promote completeness into authority

`moduleDescriptorDeclarationsComplete()` reports only whether every structural declaration required by the descriptor contract is present. Even a true result is not E1 admission, authorization to power, specimen conformance, or bench acceptance. Exact identity, primary sources, traced topology, ratings, measurements, and safe-state evidence remain open.

Vocabulary discipline. Say “declaration complete enough for E1 review,” never “this module is safe,” “qualified,” or “supported.”

9 Pass every structural gate

`validateModuleThresholdDescriptor()` checks encoding, identity and revisions, ordered ranges, topology-specific output and pull fields, declared durations, and threshold-control semantics. It is stateless and repairs nothing.



Alternative text: five hand-drawn sieves check identity and revisions, ranges, topology and pulls, timing, and adjustment semantics; only one coherent descriptor exits while malformed cards remain outside.

1. Reject invalid enum encodings or zero required identities/revisions.
2. Reject inverted millivolt ranges or raw domains.
3. Enforce topology-specific comparator, pull, and canonical-unused fields.
4. Enforce known/unknown duration canonicity.
5. Enforce fixed/potentiometer direction cross-fields.

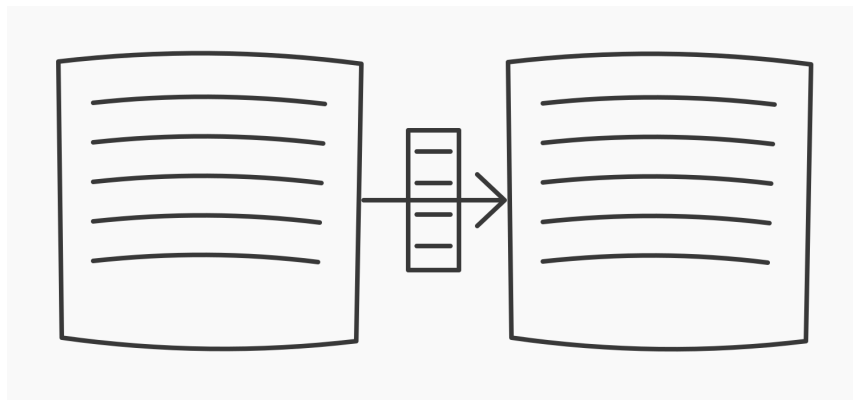
`validateModuleThresholdFrame()` next checks exact descriptor, declared-specimen/electrical provenance, source, topology, presence, statuses, producer statuses, assertion, and duration-satisfaction correlation. Invalid structure returns status; an attributable `ProducerFault` remains a valid copied frame rather than being rewritten as malformed input.

10 Exercise collisions

Predict the first disposition for each pair: invalid polarity encoding plus producer fault _____; descriptor revision mismatch plus stale analog channel _____; coherent producer fault plus an otherwise in-range value _____.

11 Compare declarations without claiming interchangeability

Two separately identified fixture revisions may share the same consumer-shaped contract while retaining different supply ranges, signal ranges, output stages, pulls, polarity, threshold controls, directions, and timings.



Alternative text: two separately identified descriptor cards have the same outer consumer shape but retain visibly different revision, polarity, pull, timing, and range marks.

Semantic equality is field-wise. A raw-object `memcmp` can inspect padding and compiler representation that the public contract never assigned meaning. Lesson 070 has no codec or persistence claim. Lesson 072 later owns a fixed characterization-record image that must bind the relevant complete declaration and evidence.

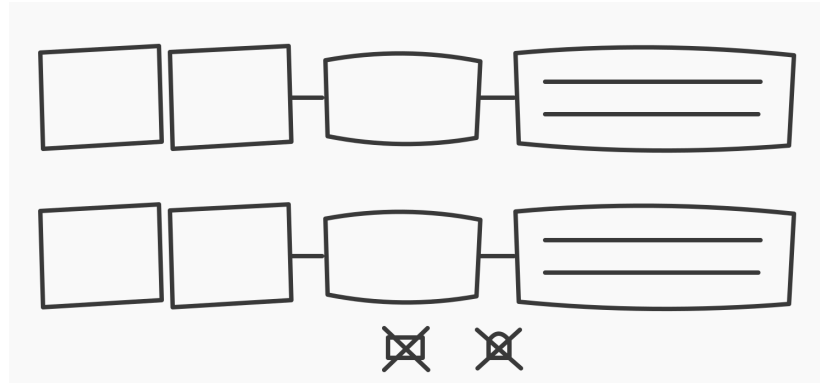
Tempting shortcut	Required correction
same seller alias	preserve distinct declared specimen references
same connector count	retain exact pinout/topology as future E1 evidence
same threshold behavior	retain polarity, pull, range, timing, and revisions
complete declaration	still require physical identity and bench review

12 Keep earlier lesson meaning intact

Lesson 070 does not retrofit module identity into `AnalogInput`, `DigitalInput`, or `ThresholdInput`. It does not reinterpret the E0 quality classifications in Lessons 034–042 or 061–063 as electrical qualification. Lesson 071 owns empirical supplied-sweep characterization.

13 Replay the stateless contract

The canonical Mega sketch acquires copied descriptor and frame fixtures, configures their declared identity, starts no hardware, validates them, and publishes named volatile topology, polarity, raw, comparator assertion, channel-status, duration-satisfaction, provenance, declaration-completeness, and returned-status cells.



Alternative text: identical descriptor and frame card pairs pass through two stateless validation runs and produce matching named output cells; crossed-out power and wiring marks remain outside the replay.

1. **Predict:** write the expected topology, polarity, comparator assertion, channel statuses, duration flags, declaration-completeness result, and returned statuses.
2. **Observe:** compile for Mega 2560 and inspect named volatile cells with a debugger, simulator, or disassembly. Serial is optional support.
3. **Interpret:** compare two runs field by field. Structural agreement proves deterministic copied behavior, not fixture truth.

The named result cells are the E0 non-Serial observation path. E0 claims zero pins, ADC channels/references, timers, interrupts, buses, rails, endpoints, power resources, or registry entries. Therefore resource acquisition and electrical safe-state evidence remain distinct open E1 checks.

14 Deterministic test matrix

Cover every valid and invalid enum encoding; topology/output/pull/rail/polarity/ control/direction cross-product; zero, one, and maximum identities/revisions; equal, ordered, and inverted ranges; known-zero, known-nonzero, and unknown durations; all three channel topologies; active-high/low assertion; producer fault collisions; provenance drift; canonical absent fields; and proof that no accepted or rejected fixture invokes an endpoint, clock, registry, transport, or power operation.

15 Troubleshoot without inventing hardware facts

Symptom	Inspect	Interpretation
descriptor invalid	enum, identity, revision, ranges	malformed declaration
frame invalid	descriptor and declared-fixture provenance	copied evidence does not correlate
assertion fails	comparator polarity	unspecified never chooses a default
channel producer fault	producer status and channel status	attributable valid fault evidence
unknown duration marked satisfied	declaration and copied flag	noncanonical frame
completeness false	every required declaration field	not ready even for E1 review planning

16 Exercises

1. Construct an analog-only descriptor and list every comparator field's canonical unused value.
2. For one high copied comparator level, predict assertion under all three polarity values.
3. Explain why `OpenDrain` plus `PullUp` still cannot choose a resistor value or rail implementation.
4. Create two descriptors with identical behavior but different declared specimen references. Explain why they are unequal.
5. Explain why true declaration completeness is not physical admissibility.

17 No circuit at E0

There is no pin-by-pin connection list, wiring diagram, or formal schematic because Lesson 070 has no circuit. Never connect an unidentified module. Future E1 work must record exact chip and PCB markings, front/back photographs, primary sources or a traced circuit, connector pin order, supply and logic limits, regulator, comparator stage, pull rail and resistance, onboard LED loading, adjustment direction, warm-up and settling measurements, ADC reference/source impedance, current, named analog/digital/power/ground test points, inactive default, rollback, power removal, and bench observations.

This lesson admits no register device, unidentified emitter, gas exposure or heater, concentration or alarm claim, physiological claim, mains, ignition, laser, pyrotechnics, generic analog-temperature board, generic capacitive-touch board, or family absent from the authorized inventory.

18 Open acceptance record

Host evidence	Record strict tests, no-exception tests, standalone header, closed-enum matrix, correlation failures, and zero hardware-call proof.
Mega evidence	Record canonical compile-only replay flash/static SRAM and named-cell inspection; this is not a powered-module result.
Memory evidence	Exact no-LTO probe: 4,546 B flash, 690 B static SRAM, 75 B synchronous stack, 45 B descriptor, 38 B frame, and 83 B descriptor-plus-frame composition; all target and hard gates pass.
Document evidence	Two-pass build, metadata, font embedding, text extraction, monochrome, visual classification, links, 200% zoom, and rendered-pencil review must pass.
Physical evidence	Open E1: exact identity, declaration truth, voltage, topology, pins, timing, adjustment, acquisition, safe state, and bench acceptance remain unproved.

Acceptance item	Record
Commit and toolchain	
Host / sanitizer / enum matrix	
Mega flash / static SRAM / stack	
Descriptor / frame sizes	
Named replay cells	
PDF and HTML accessibility review	
Exact E1 specimen owner / next action	

Stop condition. Call the result a structurally accepted copied declaration and frame. Do not call it a safe, qualified, admissible, supported, or hardware-verified module. Declaration completeness merely opens a later evidence review.

Next: Lesson 071 — Characterize one declared fixture with supplied sweeps.